

# EXECUTIVE SUMMARY

## 1. PROBLEM STATEMENT AND TARGET USERS

In home and healthcare environments, elderly individuals, post-operative patients, and people with limited mobility often require assistance for basic daily needs.

Existing systems typically use a single alert mechanism (like a buzzer), which does not clearly indicate the user's specific requirement. This leads to:

- Delayed response
- Miscommunication
- Increased dependency on caregivers

This problem becomes more critical in home settings where continuous monitoring is limited.

### Target Users:

- Elderly individuals
- Patients in recovery
- Caregivers

## 2. SOLUTION OVERVIEW

The proposed system is a smart assistance device designed for simple and clear communication using audio output (speaker).

### Key Features:

- Button-based input:
  - Food
  - Water
  - Restroom
  - Emergency
- Voice input using microphone
- Embedded ML-based command recognition
- Speaker output for clear message delivery

**Functionality:**

- Detects user input (button or voice)
- Identifies the request using ML
- Plays a predefined voice message through speaker (e.g., “Need Water”, “Emergency Help Required”)

This ensures the caregiver immediately understands the exact need without confusion.

### 3. SYSTEM DESIGN APPROACH (EDGE PROCESSING)

The system processes all inputs locally (edge-based) without relying on internet connectivity.

**Advantages:**

- Instant response
- Works offline
- Low system complexity

### 4. Why MG26 + SiWx917

**MG26:** ML processing, GPIO handling, real-time control, low-power operation

**SiWx917:** Wi-Fi/Bluetooth connectivity, data communication, cloud/interface support

### 5. KEY DIFFERENTIATION AND EXPECTED IMPACT

**Key Differentiation:**

- Combines button + voice input
- Provides clear spoken output instead of vague alerts
- Fully offline operation
- Easy to use for elderly users

**Expected Impact:**

- Faster caregiver response
- Reduced confusion and repetition
- Improved quality of care
- Increased independence for users